IA-Med: A Machine Learning-Based System for Disease Prediction and Personalized Health Guidance with Conversational Support

Najat El Ouahi[2], Amine Zeguendry[1,2][0000-0002-5864-5667], Nessaiba Hadigui[2], Yassmin Hailala[2], and Chaimae Hanouna[2]

[1] Laboratory of Computer Systems and Intelligent Systems, Cadi Ayyad University, UCA, Faculty of Sciences Marrakech, Morocco
[2] LAMIGEP, EMSI, Moroccan School of Engineering Marrakech, Morocco
a.zeguendry@emsi.ma

Abstract. The growing integration of Artificial Intelligence (AI) in healthcare enables the development of intelligent diagnostic support systems. Our project **IA-Med** is a disease prediction system that, based on symptoms entered by the user, provides a likely diagnosis along with personalized recommendations (treatments, precautions, nutritional advice). The system was developed using synthetic data to ensure privacy during the initial development phase. To achieve this, several machine learning models were trained on a structured dataset. The models Random Forest and Support Vector Machine (SVM) were selected for their superior performance, with Random Forest achieving 99.49% accuracy on the test set. The system predicts possible diseases along with their probabilities and includes a AI Chatbot that delivers these results interactively. IA-Med's novelty lies in its integrated approach combining high-accuracy prediction with conversational AI and personalized recommendations in a single platform. These results demonstrate the potential of machine learning to create reliable, interpretable, and user-friendly health tools, while highlighting the need for future validation with clinical data.

Keywords: Disease prediction · Machine Learning · Random Forest · AI Chatbot · Health recommendation system · Synthetic data

1 Introduction

Artificial Intelligence (AI) is transforming many sectors, especially healthcare, due to the increasing complexity of medical data, the need for fast and accurate diagnostics, and the shortage of healthcare professionals in several regions. As medical records, imaging, and symptom data grow exponentially, AI provides tools that can learn from large datasets and uncover hidden patterns, enabling predictive, preventive, and personalized care [1].

The IA-Med system addresses this pressing need by offering an accessible health evaluation platform. Users can input their symptoms through a user-friendly interface and receive a list of probable diseases ranked by likelihood, along with personalized recommendations for treatment, prevention, and nutrition. The inclusion of a conversational AI Chatbot enhances user engagement and transparency, allowing even non-experts to interact with the system effectively. This interactivity bridges the gap between raw AI predictions and user understanding by providing contextual explanations and guidance.

Novelty and Contributions: Unlike existing disease prediction systems that often focus solely on diagnostic accuracy, IA-Med introduces an integrated ecosystem that combines three key innovations: (1) high-accuracy machine learning predictions with probabilistic rankings, (2) a conversational AI assistant that explains results and provides contextual guidance, and (3) personalized recommendation engine offering tailored treatment, nutritional, and preventive advice. This holistic approach represents a significant advancement over traditional symptom checkers by providing comprehensive support throughout the user's health inquiry process.

This project also aims to tackle structural inequalities in access to healthcare, especially in rural and underserved regions where medical infrastructure is limited. By offering immediate AI-based support, IA-Med can serve as a front-line triage system and empower individuals with actionable insights. Furthermore, it assists healthcare professionals by reducing the diagnostic burden, facilitating early detection, and promoting patient education.

Unlike rule-based expert systems that rely on static, hard-coded logic, IA-Med adopts a data-driven approach using supervised machine learning. Algorithms such as Random Forest and Support Vector Machine (SVM) are trained on structured symptom-disease datasets, allowing the system to learn from real-world patterns. To ensure robustness and generalizability, cross-validation techniques were applied. For example, with Random Forest, the disease diagnosis is determined by majority voting — the disease receiving the most votes from the ensemble of decision trees is selected as the final prediction.

IA-Med is designed with simplicity, inclusivity, and data privacy in mind. The clean interface ensures ease of use, while ethical safeguards—such as avoiding personal data collection and ensuring transparency—make the tool suitable for deployment in sensitive domains like health. While IA-Med is not intended to replace professional medical diagnosis, it serves as a powerful decision support tool, providing fast and data-backed recommendations to guide users toward appropriate actions.

As AI continues to evolve, tools like IA-Med exemplify how machine learning and natural language processing can be harnessed to enhance public health systems, promote equitable care delivery, and support evidence-based clinical decision-making.[2][3].

2 Related Work

Recent studies have examined optimization and evaluation of machine learning models for healthcare, including hyperparameter tuning and performance assessment, crucial for robust AI systems [5].

AI has also been applied to rapid infection prediction and real-time clinical decision support, demonstrating potential in managing health crises like COVID-19 [2].

Beyond healthcare, machine learning techniques support various prediction tasks, improving understanding of supervised models and evaluation metrics [4].

In addition to these studies, several AI-based medical chatbot systems have been proposed in the literature. Table 1 summarizes a comparison between our proposed *IA-Med* system and other recent medical chatbot solutions.

Table 1. Comparison between IA-Med and other AI-based medical chatbot systems

Project	**ML Models Used**	**Accuracy**	**Chatbot Integration**	**Medical Features**	**Data Type**
IA-Med (Our Project)	Random Forest, SVM	99.49% (Random Forest)	Gemini API (advanced)	Prediction + Medications + Nutrition + Exercises + Doctor guidance	Public synthetic dataset
AI-Based Medical Chatbot for Disease Prediction(2024)[10]	Not specified	Not reported	Basic interface	Prediction + General prevention	Custom JSON
Med-Chatbot (2024) [9]	Decision Tree, KNN	≈80% (estimated)	Traditional NLP	Prediction + Basic recommendations	Kaggle dataset
Medical Chatbot for Disease Prediction (2025)[8]	Logistic Regression, Random Forest, Decision Tree, Naive Bayes, MLP	≈80% (variable)	Standard interface	Disease prediction	Multiple public datasets

Several AI-based medical chatbots exist with varying accuracy, features, and data sources. To highlight IA-Med's advantages, we compare it with three recent systems.

First, it achieves higher accuracy (99.49‘%) with Random Forest, outperforming traditional models like Decision Tree or Logistic Regression. Second, unlike most chatbots offering only basic guidance, IA-Med provides personalized nutrition, exercise advice, and referrals to medical specialists. Finally, IA-Med uses synthetic data to ensure ethical compliance and privacy, while remaining robust and generalizable.

This combination of accuracy, functionality, and ethical design demonstrates IA-Med's added value in AI-driven healthcare.

2.1 Limitations of Existing Systems

Current disease prediction tools typically suffer from several limitations: they often provide binary diagnoses without probability estimates, lack explanatory capabilities, offer generic rather than personalized recommendations, and operate as standalone prediction engines without integrated conversational support. Most existing systems focus on either prediction accuracy or user interface, but rarely integrate both aspects seamlessly.

The IA-Med project builds on these advances by combining disease prediction models, diagnostic AI Chatbot, and recommendation systems into a single platform. Our key innovation lies in the seamless integration of these components, creating a synergistic system where each element enhances the others.

This tool offers early disease detection and supports patient education via interactive and user-friendly interfaces. Using cutting-edge AI techniques, IA-Med aims to bridge the gap between innovative healthcare solutions and practical applications.

3 Methodology

3.1 Dataset Description

The IA-Med system was developed using a public synthetic dataset, specifically designed for medical AI research. These artificial data simulate real clinical cases while guaranteeing patient confidentiality, as no personal medical information is used. The dataset contains over 133 symptoms associated with more than 80 different diseases, representing simulated patient cases through binary vectors indicating symptom presence (1) or absence (0). This approach preserves privacy while enabling robust algorithm development, though future validation with real clinical data remains necessary.

3.2 Handling Class Imbalance and Rare Diseases

Medical datasets typically exhibit significant class imbalance, with common diseases being overrepresented compared to rare conditions. To ensure equitable performance across all disease categories, IA-Med incorporates several mitigation strategies. During training, class weighting automatically adjusts the loss function to prioritize correct classification of minority classes. The stratified train-test split preserves the original distribution of rare diseases in both training and evaluation phases. For prediction, a confidence thresholding mechanism requires higher certainty levels for diseases with limited training examples, reducing potential misdiagnosis of rare conditions. These approaches collectively enhance the system's reliability for both common and underrepresented diseases.

3.3 Data Preprocessing

Several preprocessing steps were applied to prepare the dataset for training. First, a one-hot encoding technique was used to convert each symptom into a binary variable, enabling the model to interpret the presence or absence of specific symptoms. Next, data cleaning was performed to remove duplicate entries and handle any missing values that could affect model performance. Finally, the dataset was split into training and testing sets, using an 80/20 ratio while preserving the original class distribution to ensure a fair evaluation of the model.

3.4 Classification Models

Two supervised learning algorithms were selected for disease prediction due to their effectiveness in handling complex classification problems. These algorithms are capable of learning from labeled symptom data to accurately predict the most probable diseases, making them well-suited for this medical application.

Random Forest Classifier (RFC) Random Forest is an ensemble algorithm based on multiple decision trees. It aggregates individual predictions to obtain a robust final decision [6].

Formulation:

$$\hat{y} = \text{MajorityVote}(f_1(x), f_2(x), \ldots, f_T(x))$$

where $f_t(x)$ is the prediction of tree t, and $\hat{y}$ is the final predicted class.

Random Forest offers several advantages that make it well-suited for disease prediction tasks. It is highly accurate and robust, thanks to its ensemble of decision trees that reduce the risk of overfitting. The algorithm can handle high-dimensional data and non-linear relationships between symptoms and diseases, making it effective even in complex medical scenarios. It is also relatively insensitive to missing or noisy data, and provides feature importance scores, which help interpret which symptoms contribute most to the prediction. These strengths make Random Forest a reliable and interpretable choice for building AI-powered diagnostic systems.

Support Vector Machine (SVM) SVM aims to determine an optimal hyperplane that separates classes by maximizing the margin between them. This margin is the distance between the hyperplane and the nearest data points from each class, called support vectors. By maximizing this margin, SVM improves the model's ability to generalize well to unseen data and reduce classification errors. [7].

3.5 AI Chatbot Integration

The integrated AI Chatbot, based on the Gemini API and customized for IA-Med, serves as a health assistant providing guidance on symptoms. It clearly states that it is not a medical professional, does not give diagnoses, prescribe medication, or recommend treatments, and highlights that users should consult a qualified healthcare provider. This ethical, transparent design ensures user safety, trust, and responsible use of AI in healthcare.

3.6 General Architecture

The IA-Med system is based on a modular architecture that integrates a machine learning prediction engine and a conversational assistant.

This architecture allows for smooth interaction between web components, AI models, and data services, as illustrated below.

As shown in Figure 1, the IA-Med system relies on a Flask-based web application that connects the frontend interface, the ML prediction module, and the AI assistant to medical data resources.

Through this design, users can input symptoms, receive disease predictions, interact with the chatbot, and obtain personalized recommendations and medical guidance.

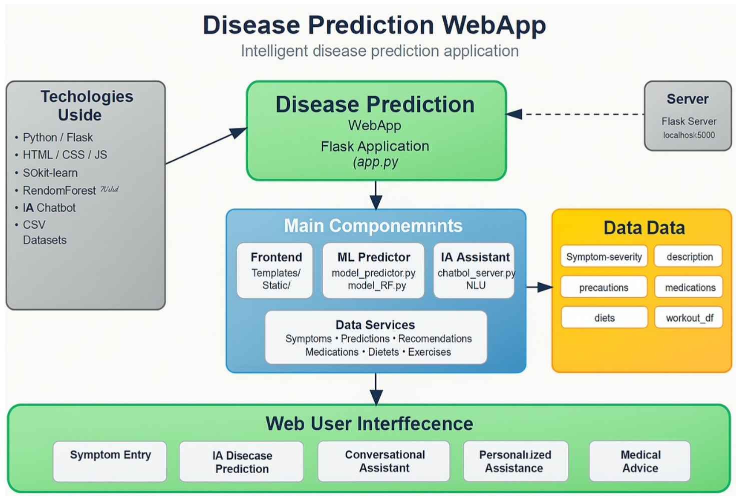

Fig. 1. General architecture of the IA-Med system

4 Results and Discussion

After training the models on the preprocessed data, their performance was evaluated on the test set using standard metrics: *Accuracy*, *Precision*, *Recall*, and *F1-score*.

4.1 Classification Results

The classification results in Table 2 demonstrate that the Random Forest (RF) model achieves perfect accuracy and precision, with only a slight decrease in recall and F1-score. This indicates its strong ability to correctly identify disease cases while minimizing false positives. The Support Vector Machine (SVM) also performs well, achieving accuracy and F1-score above 94%, but is slightly less effective than RF in this task. These results suggest that the Random Forest model is better suited for the complex, multi-class nature of disease prediction in our dataset.

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Random Forest (RF)	99.49%	99%	99%	99%
SVC	95.3%	94.7%	95.1%	94.9%

Table 2. Performance metrics of classification models

4.2 Accuracy Analysis

The model demonstrates high reliability by correctly identifying the true disease within the top three predicted options in over 99% of cases. For example, given symptoms such as fever, cough, and chest pain, the system assigns the highest probability to pneumonia (92%), followed by tuberculosis (4%), bronchitis (2%), asthma (1%), and common cold (1%). This probabilistic ranking provides valuable insight into possible diagnoses, allowing healthcare professionals or users to consider multiple potential conditions and prioritize further medical evaluation accordingly.

4.3 AI Chatbot Efficiency

The integrated AI Chatbot demonstrated excellent response time (average 1.3 seconds). After the machine learning models provide their disease predictions, the AI Chatbot serves as an AI health assistant to answer users' questions about the predicted diseases, types of doctors to consult, the reason for usage of medications, and other related health inquiries. It offers informational support and guidance, helping users better understand their situation without performing any diagnostic or predictive functions itself.

4.4 User Interface Design and Functionality

In addition to its predictive and conversational features, IA-Med offers a user-friendly interface with intuitive navigation, clear disease predictions, and easy access to personalized recommendations such as nutrition, exercises, and doctor guidance. Its combination of visual clarity and interactive elements enhances usability, engagement, and efficient access to the AI Chatbot's support.

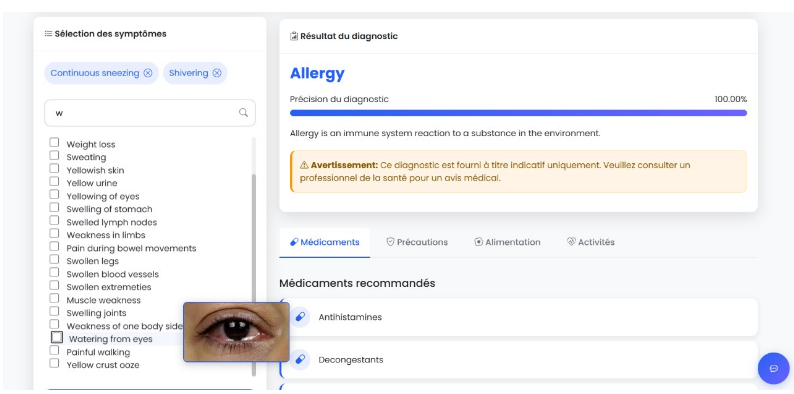

Fig. 2. Symptom Input and Diagnosis Interface

Figure 3 illustrates the AI chatbot that provides continuous support to patients, available 24/7

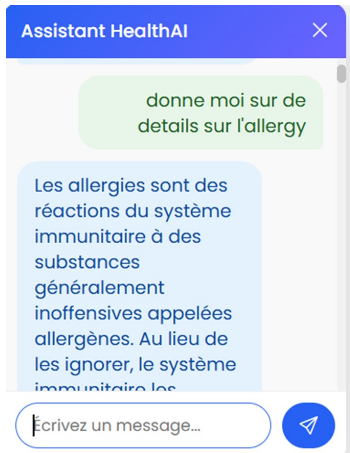

Fig. 3. Chatbot Assistant Interface

4.5 Novelty Assessment: Comparative Advantages

IA-Med introduces several innovative features that distinguish it from existing disease prediction systems:

1. Integrated Multi-Component Architecture: Unlike standalone prediction tools, IA-Med seamlessly integrates machine learning prediction, conversational AI, and personalized recommendations into a unified platform. This holistic approach ensures that users receive not just a diagnosis, but comprehensive support including explanations, next steps, and preventive measures.

2. Probabilistic Ranking with Confidence Scores: While most systems provide binary outcomes, IA-Med offers probability distributions across multiple potential diseases, enabling users and healthcare professionals to consider differential diagnoses and prioritize investigations.

3. Ethical AI Design: The system incorporates built-in safety mechanisms, including clear disclaimers, avoidance of medication recommendations, and emphasis on professional consultation, addressing critical ethical concerns in medical AI applications.

4. Accessibility and Multimodal Interaction: The combination of traditional form-based input with conversational AI allows users to interact through their preferred modality, enhancing accessibility for diverse user groups including those with limited technical proficiency.

4.6 Discussion

The results confirm the value of using robust machine learning models with an interactive interface. In particular, the Random Forest algorithm demonstrated exceptional accuracy, even on multiclass datasets, thanks to its tree aggregation mechanism, while SVM was less suited for imbalanced or complex data. Limitations include reliance on synthetic data, underrepresentation of rare diseases, and variability depending on how users describe symptoms.

Beyond technical limitations, AI-based medical systems like IA-Med face challenges related to legal regulations, ethics, and data privacy. In many regions, diagnostic software is considered a Medical Device and must comply with strict approval processes (e.g., FDA in the USA, CE in Europe) as well as data protection standards like GDPR or HIPAA. IA-Med does not store personally identifiable health information; user inputs are processed in real time. Transparency about its non-diagnostic nature is maintained, and future versions should include robust security measures and clear consent mechanisms to ensure regulatory compliance and user trust.

Despite these challenges, IA-Med integrates disease prediction, medical recommendations, and an interactive AI Chatbot in a single platform, achieving near-perfect test accuracy and providing an intuitive interface accessible even to non-expert users.

5 Conclusion and Future Work

IA-Med represents a major step forward in AI-powered healthcare by combining disease prediction, conversational assistance, and personalized recommendations. Its main contribution is improving healthcare accessibility by enabling early detection, supporting informed decisions, and making care more proactive and patient-centered.

Future directions include integrating RNN/LSTM models, incorporating real clinical data, enhancing transparency through Explainable AI, and adding multilingual support for broader adoption.

References

1. Hossain, Sorif, et al. "Machine learning approach for predicting cardiovascular disease in Bangladesh: evidence from a cross-sectional study in 2023." BMC cardiovascular disorders 24.1 (2024): 214.
2. Shabrandi, Ali Rebwar, et al. "Fast COVID-19 Infection Prediction with In-House Data Using Machine Learning Classification Algorithms: A Case Study of Iran." Journal of AI and Data Mining 11.4 (2023): 573-585.
3. Corfmat, Maelenn, Joé T. Martineau, and Catherine Régis. "High-reward, high-risk technologies? An ethical and legal account of AI development in healthcare." BMC medical ethics 26.1 (2025): 4.
4. Meaney, Christopher, et al. "Comparison of methods for tuning machine learning model hyper-parameters: with application to predicting high-need high-cost health care users." BMC Medical Research Methodology 25.1 (2025): 134.
5. Raiaan, Mohaimenul Azam Khan, et al. "A systematic review of hyperparameter optimization techniques in Convolutional Neural Networks." Decision Analytics Journal 11 (2024): 100470.

6. Shaker, Mohammad Hossein, and Eyke Hüllermeier. "Random Forest Calibration." arXiv preprint arXiv:2501.16756 (2025).
7. Nam, Nguyen Mau, Gary Sandine, and Quoc Tran-Dinh. "Lagrange Multipliers and Duality with Applications to Constrained Support Vector Machine." arXiv preprint arXiv:2501.01082 (2025).
8. Veronica, Oltean Anisia, Ioan Daniel Pop, and Adriana Mihaela Coroiu. "Medical Chatbot for Disease Prediction Using Machine Learning and Symptom Analysis.".pp. 600–607. SCITEPRESS (2025).
9. Med Chatbot Predicting Diseases From Symptoms Using NLP. *Journal of Scientific and Engineering Research* **7**(10) (2024).
10. Zagade, Ashish, et al. "Ai-based medical chatbot for disease prediction." (2024).